

1 Which branch of the tree moved from Q1 to Q2?

revenue Rs
2.10 to 1.87 cr

customers 69
to 69

orders per
customer
1.65 to 1.25

revenue per
order Rs 1.84
to 2.17 lakh

The six chapters climb it in order: is the drop real, which branch moved, which segment moved, did customers pay more, were customers lost, and did the button do it.

Revenue fell 11.0 percent on closed quarters: 1.000 times 0.754 times 1.180 is 0.890. In behaviour the fall sits in Retail-Plus, where the same 22 members placed 26 orders against 51.

Crux. Confirm the drop on matched windows before you explain it.

2 Is the drop real once the windows match?

Window	Result	Verdict
Q1 against Q2 cut at 15 Sep	Down 25.9 percent	First and last dates show 13 weeks against 11.
Closed quarters, 13 weeks each	Down 11.0 percent	It is fair once Q2 has closed.
Same 11 weeks, Q2 still open	Down 17.0 percent	Matched weeks while a quarter is open.
Per week on the cut window	Down 12.4 percent	A rate, said with its window.
Second route, by month	Down 11.0 percent	The dates agree with the quarter field.

Crux. A rate without its denominator is a rumour.

3 How many rupees does each branch carry?

Step	Rupees
Q1 revenue	Rs 2,10,00,000
Customers, 69 to 69	Rs 0
Orders per customer, 28 fewer orders	Minus Rs 51,57,895
Revenue per order, Rs 33,231 more on 86	Plus Rs 28,57,895
Q2 revenue	Rs 1,87,00,000

Crux. Decompose along the tree: customers, orders per customer, revenue per order.

4 What does a missing discount count as?

```
order.get("discount", 0) # silently makes a blank Rs 0
```

Read as zero, a blank counts as "no discount", so 50.0 percent of Q2's orders seem to carry one. Split the rest into recorded zeros and blanks, report the share where recorded, 71.7 percent, with the range the blanks allow, and bound the branch: Rs 150 on 86 orders is at most Rs 12,900.

Crux. Missing means unknown until someone chooses a default and writes down why.

5 How do four segments roll up into one rate?

One function replaces eight copied loops and eight places to edit. A plain average of four segments gives 1.94 and 1.82; total orders over total customers gives 1.65 and 1.25, the figure a roll-up must reproduce.

Crux. Roll a rate up with its weights; never average the averages.

6 Did customers pay more, or did the mix move?

Revenue per order rose Rs 33,231 and no segment rose 18 percent: about 69 percent is mix, since small Retail-Plus orders fell from 44.7 to 30.2 percent of orders.

Crux. A blended rate can move while no segment moves; split mix from rate.

7 Does every group come back, and was anyone lost?

```
def tree_for(rows):  
    ...  
    return {"revenue": revenue, "orders": n, "customers": c}
```

A helper that prints for large changes returns **None**, and the filter drops it: four segments go in and two come back as **None**. Check with **None in changes.values()**. The id overlap answers Marketing's churn claim: 69 in both quarters, 0 lost, 0 new.

Crux. A function returns its answer; count the groups in and the groups out.

8 Did the button do it, and what would settle it?

Test	What the file shows
Timing	Retail-Plus fell from July, before the late-August break the complaint dates.
Channel	Web, store and app all fell, so the app did not fall first and alone.
Ceiling	On the 55 days before the break the button explains at most about 4 orders; on the 37 days just before it, 7.0. Both sit far below the 25 a hurried memo charges.
Evidence to ask for	The app's reorder logs by week, the release date, and the tier's change log.

Crux. Name the cause as a hypothesis, with its ceiling and the evidence that would settle it.

THE WORDS ON THIS SHEET

- Matched windows** Two periods of the same length compared, or both turned into a rate

Rate A numerator over a denominator in a stated window
- Like-for-like** A comparison that holds the outlets, the weeks and the weekends equal

Decomposition A change split along the tree into branches that multiply
- Bridge** A change in rupees moved one branch at a time from start to end

Range The largest value less the smallest, set by two orders alone
- Mix** Each segment's share of orders

Rate part The change inside segments, mix held still